



MASTER IN ASTROPHYSICS
AND SPACE SCIENCE



TOR VERGATA
UNIVERSITÀ DEGLI STUDI DI ROMA

Optimisation & Monte Carlo Sampling

INTRODUCTION TO NUMERICAL METHODS FOR ASTROPHYSICS

Fritz Ali Agildere

November 27, 2025

Overview

Probability Refresher

Definitions & Notation

Random Numbers

Optimisation

Common Methods

Problems

Markov Chain Monte Carlo

Popular Algorithms

Tuning & Diagnosing

Alternatives

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Frequentist View

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$$P(A) = \frac{k}{n}$$

- *Without Prior Knowledge*: out of n independent measurements the outcome A occurs k times, defining

$$P(A) = \lim_{n \rightarrow \infty} \frac{k}{n}$$

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Bayes' Theorem

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

Continuous Distributions

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■ *Mean* or first raw moment $\mu = E[X] = \int_{-\infty}^{+\infty} x f(x) dx$

■ *Variance* or second raw moment $\sigma^2 = E[(X - E[X])^2] = \int_{-\infty}^{+\infty} (x - E[x])^2 f(x) dx$



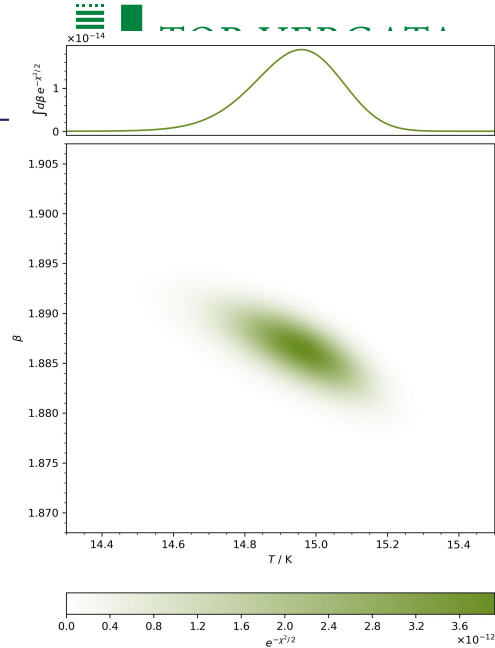
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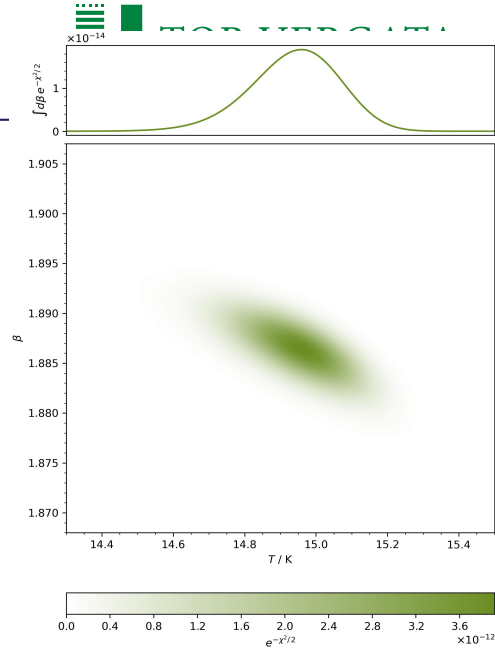
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Where I Studied

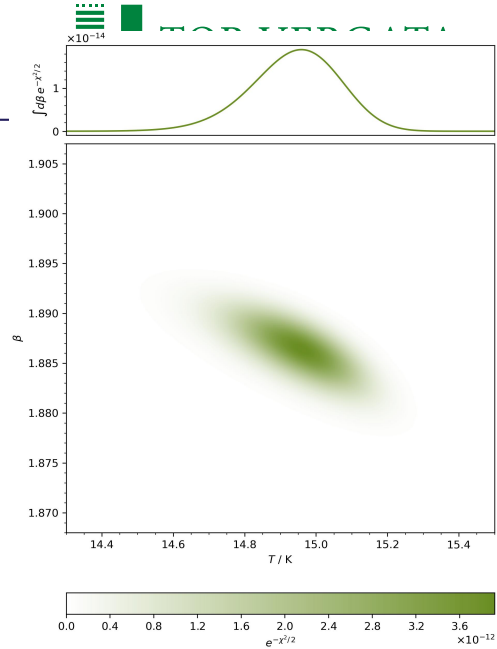
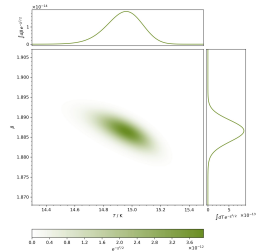
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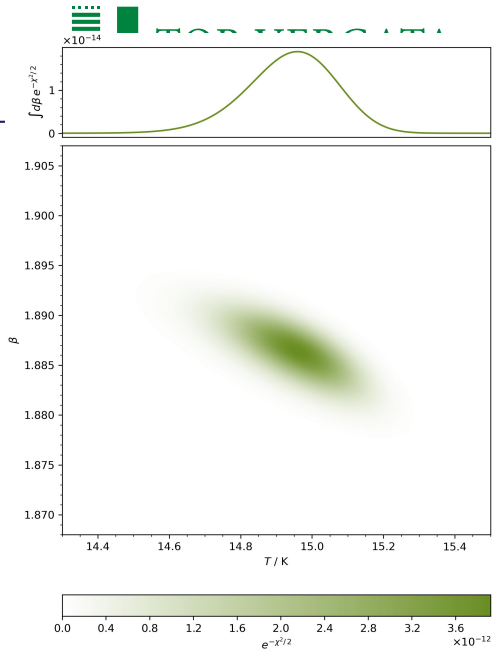
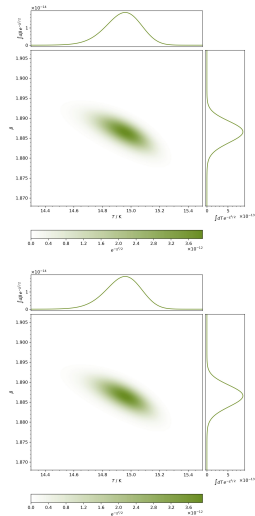
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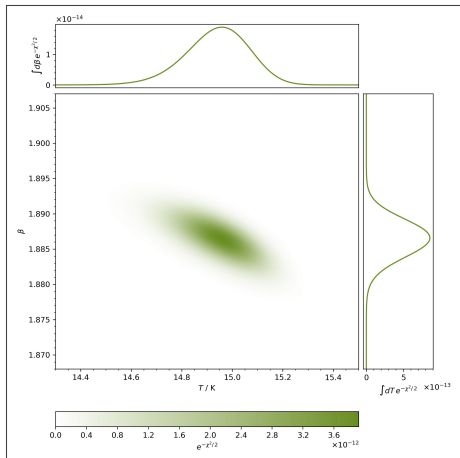
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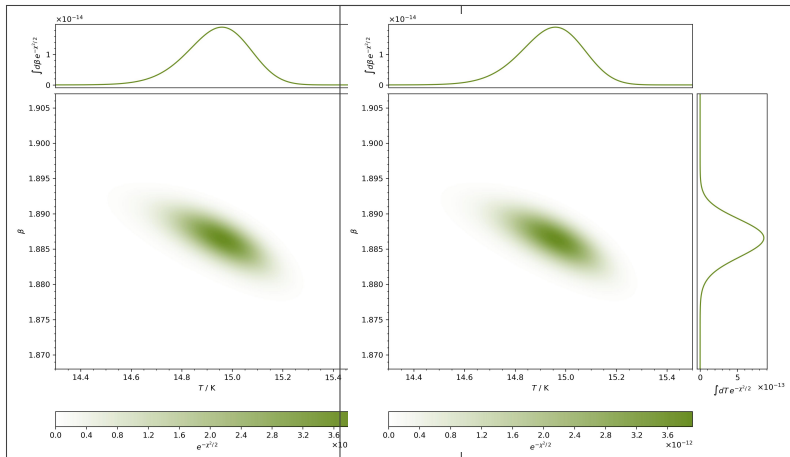
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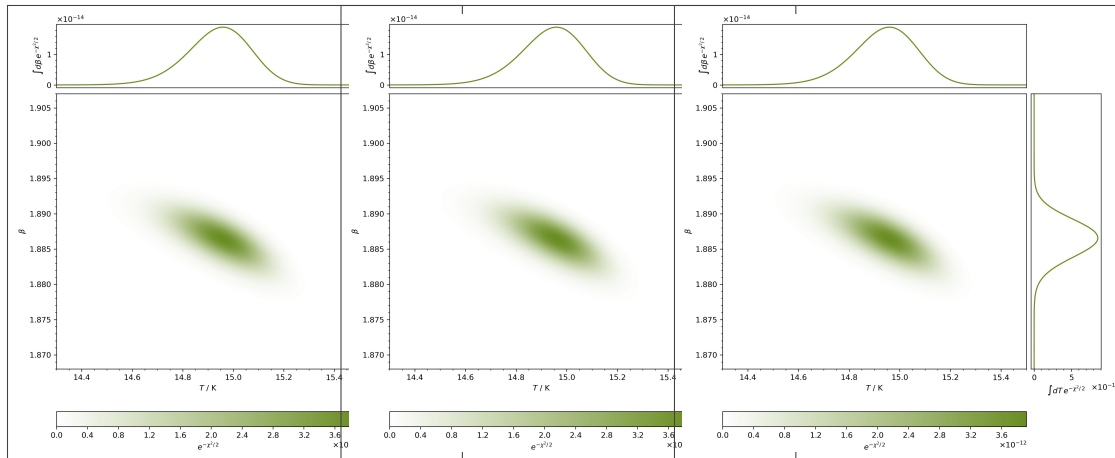
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Why Astrophysics

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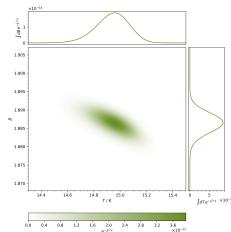
- Interesting

Why Astrophysics

- Interesting
 - Awesome Scales

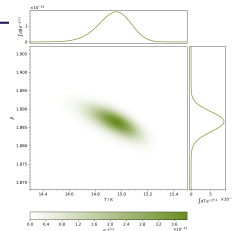
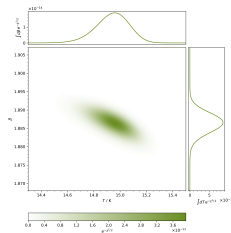
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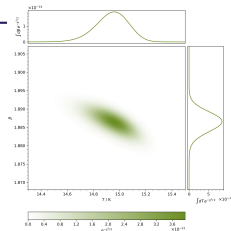
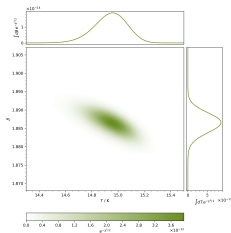
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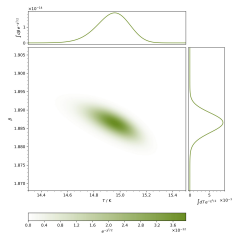
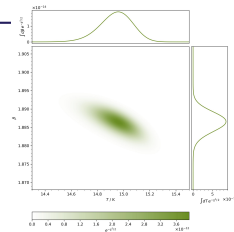
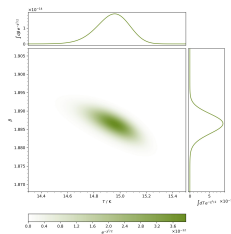
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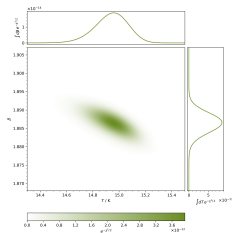
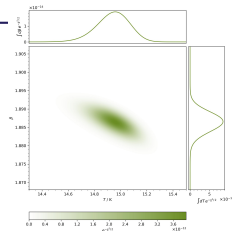
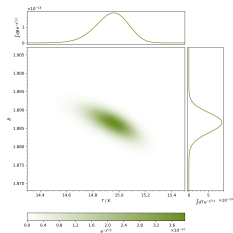
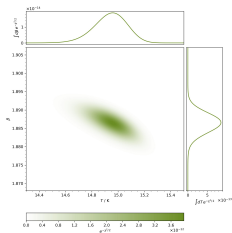
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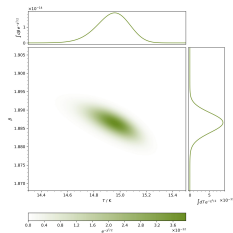
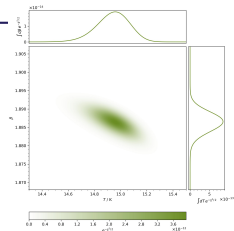
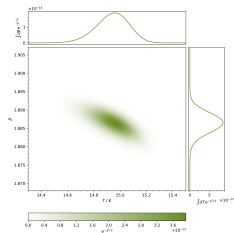
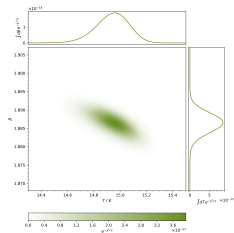
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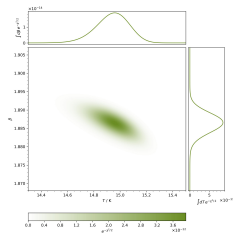
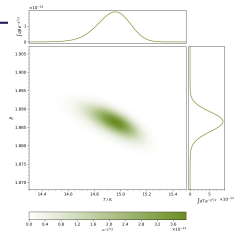
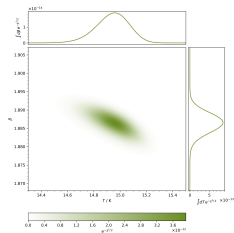
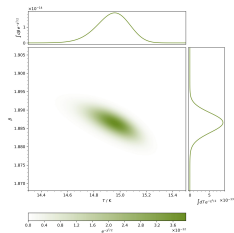
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 - Awesome Scales
 - Readily Observable
- Interdisciplinary



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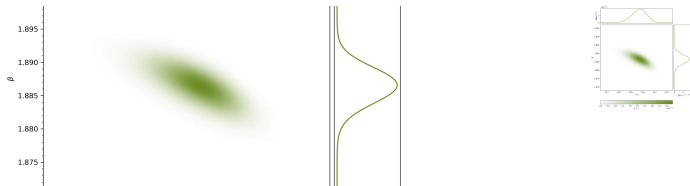
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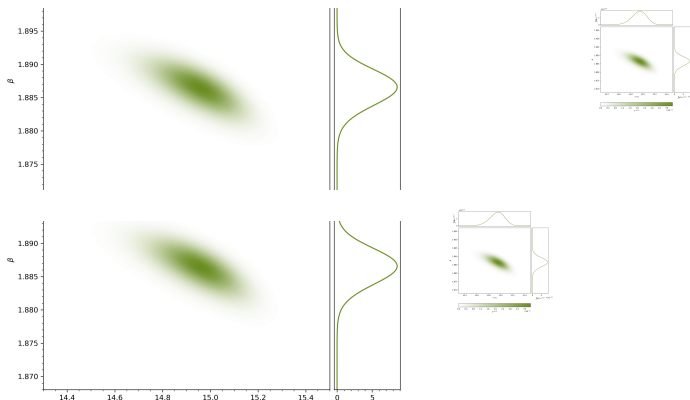
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Where I Have Been

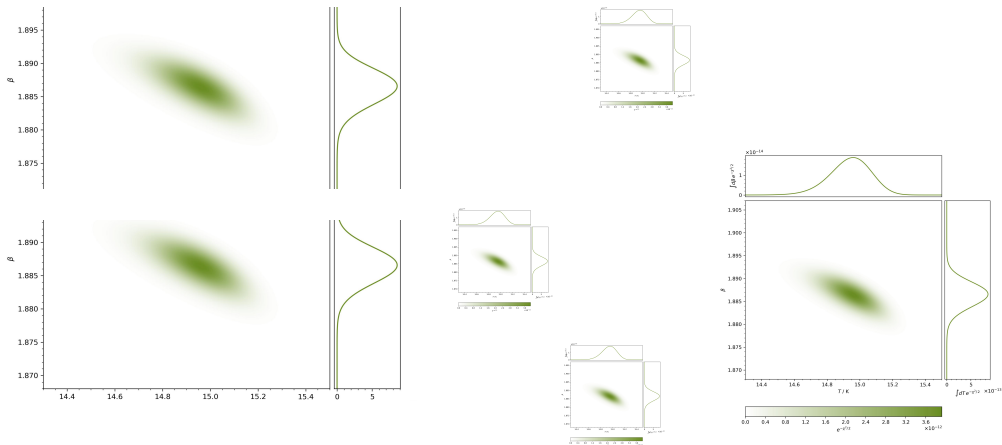
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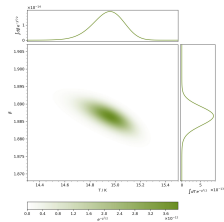
Other Interests

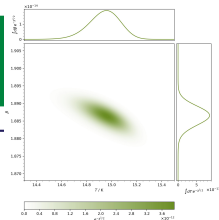
Other Interests

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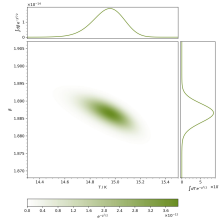
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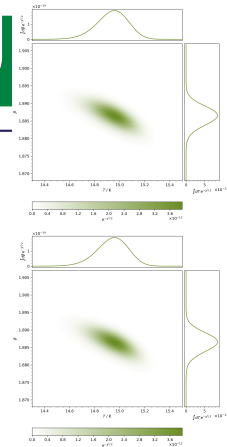
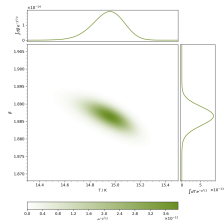
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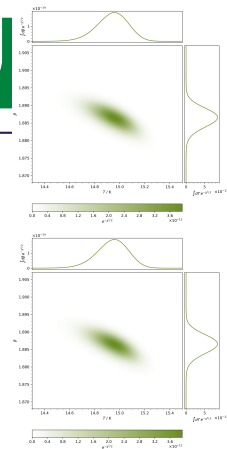
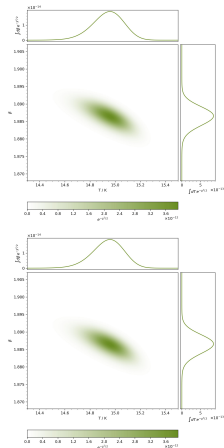
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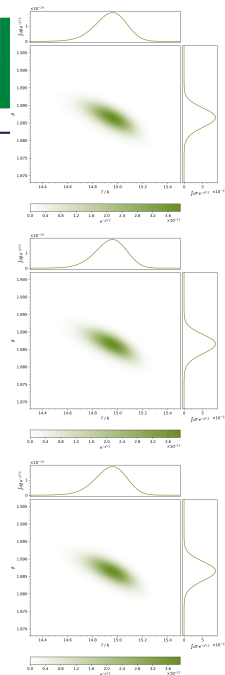
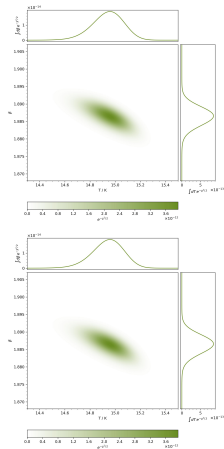
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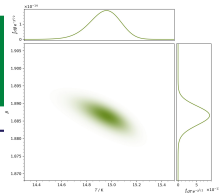
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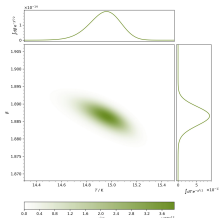
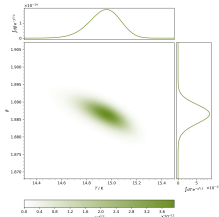
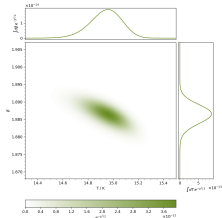
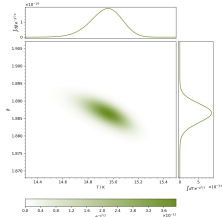
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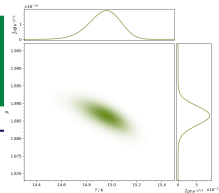




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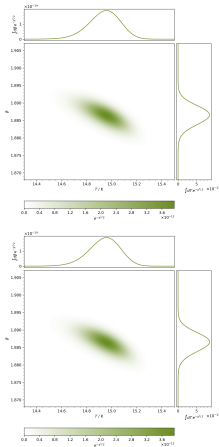
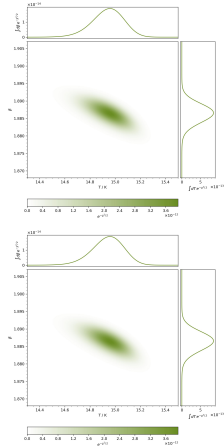
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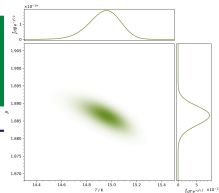




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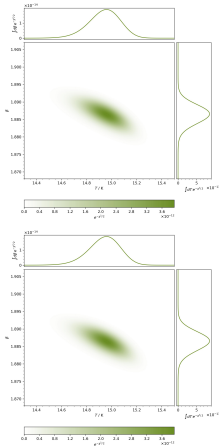
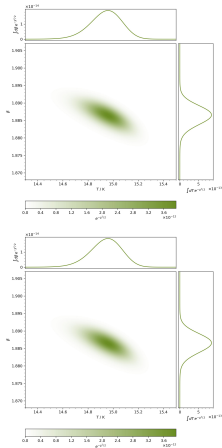
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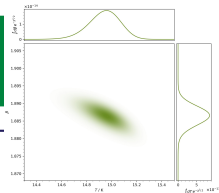




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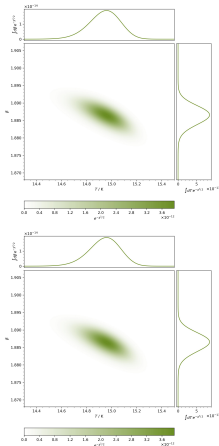
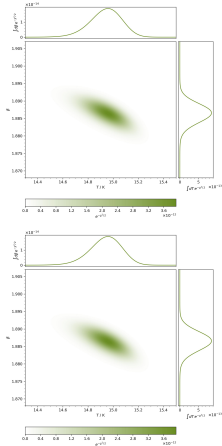
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- Programming
- Politics





Other Interests

- Nature Photography
- Hiking & Cycling
- Programming
- Politics
- Meeting New People



Thank You!